

ATLAS EDGE-SEARCH CAMPAIGN — FINAL REPORT

Pen-drive migration verification + multi-track edge search · 2026-08-11 · E:\forex-data (drive) · Account: \$150 T212 CFD

Executive summary

Total candidates tested this session: 4 spreads (full four-stage pipeline, 1000-perm MC) + 2 validation modules (vol forecast). Tracks B (literature) and C (platforms) yielded zero testable candidates after the eligibility filter — an honest, expected outcome given the project's graveyard and the drive's data constraints. All four structural spreads failed: the cointegration is real and stable out-of-sample for every spread, but the reversion payoff is 6–12× below retail CFD costs and not significant even gross (t 0.3–1.4). The volatility module **validates** (monotonic calibration OOS, permutation p<0.001) but **does not help** the one surviving strategy (the USDJPY news drift) — its edge is concentrated in high-vol states, so inverse-vol sizing hurts (Sharpe 1.73 → 1.56). Salvage attempts: 0 attempted — classified as structural, see per-candidate records.

Time budget — actual vs planned

Phase	Planned	Actual (approx.)	Note
Part 1 migration verification	15–30 min	~45 min	Engine missing from drive; archive workaround (drive writes ~50 K
Part 5.0 timing test + Track A	2.5–3 h	~25 min	Crush timing test: 9.9 s/candidate → all 4 spread at full rigor (~45 s
Tracks B + C (search)	within budget	~30 min	Zero eligible candidates, documented
Part 9.2 vol module	within budget	~25 min	Validates; does not help the drift
Part 9.4 PDF	20–30 min	~15 min	This document

Part 1 — Migration verification (all passed)

Structure: all required trees present on E:\forex-data except atlas-engine (found only on local disk). **Completeness:** events.parquet 84k+ rows 2015→present; strategies registry 26 = 26 JSONs; normalized store + manifest present. **Regressions (all three reproduce exactly):** ① hog August roll-check — continuous −13.6% (t −7.5) vs real path −0.03% (t −0.02) ■; ② Dual Momentum — holdout +2.12%/mo, t 4.51, wf p 0.001, 6/6 gates ■; ③ USDJPY news drift — IS t 1.139, OOS t 3.775, boot P(≤0) 0.0014, 4/6 gates ■. **Live:** forward ledger live (4 events, 3/4 hits); task scripts location-relative. **Issues found & fixed:** atlas-engine absent → shipped as sha256-verified archive (atlas-engine.tar.gz, 1.8 MB, byte-identical both sides); partial copy renamed atlas-engine.partial (do not use). Drive write throughput measured ~50 KB/s — the reason for archive-instead-of-per-file migration. Stray cp/tar processes were holding the drive; killed. No data loss.

Part 3 — Account constraints as used

Constraint	Value used	Where applied
Capital	\$150 total; free CFD margin not programmatically confirmed	margin discussion
Leverage caps	FX majors 30:1, commodities 10:1 (T212)	margin capacity
Spread	0.30%/leg round trip (corn 0.27%, gasoline 0.46% measured)	Track A posts
Financing	0.01%/day/leg assumption (T212 publishes no historical swapable)	Track A posts — flagged
Cross-currency fee	0.5% on P&L for non-USD-base pairs — none of Track A legs	leg affected (all USD-priced)

Track A — Structural spreads: 4 candidates, full records

Method: log-ratio spread, 90-day rolling z-score, enter $|z| \geq 2$, exit $z=0$ or 20-day stop, IS 2000-09→2014-12 / OOS 2015→2026-08. Engle-Granger cointegration (ADF, MacKinnon 5% crit -2.86) tested IS and re-tested OOS separately. Permutation blocks: 20-day joint block bootstrap (preserves cross-leg correlation). 1000 permutations at both MC stages (timing test: 9.9 s/candidate).

Spread	Coint IS (ADF)	Coint OOS (ADF)	Gross %/yr	Gross t	Net %/yr OOS	Net t OOS	WF Sharpe/t	Verdict
Soybean crush	-5.71 ■	-4.34 ■	+2.12%	1.39	-10.24%	-6.23	-0.83 / -3.2	FAIL — Stage 1 gross
Hog-corn	-3.85 ■	-5.11 ■	+1.36%	0.31	-10.83%	-2.44	-0.88 / -3.4	FAIL — Stage 1 gross
Live/Feeder cattle	-4.34 ■	-3.68 ■	+1.95%	1.2	-11.04%	-6.29	-1.67 / -6.4	FAIL — Stage 1 gross
3:2:1 Crack	-4.34 ■	-4.84 ■	+2.6%	0.62	-8.62%	-2.03	-0.48 / -1.8	FAIL — Stage 1 gross

Per-candidate detail

Soybean crush — processor margin: crusher must buy beans and sell meal+oil on the crop calendar

IS (2015-09→2014): mean -3.952%/yr, Sharpe -0.5, win 22.0%, t -1.87 — **fails Stage 1 even gross** (gross OOS Sharpe 0.09–0.41, gross t 0.31–1.39, not significant). Permutation p (S2) = 1.0 (expected: real IS result is negative). Walk-forward (S3): Sharpe -0.83, t -3.15. Roll probe: 52 candidate roll days; excluding them changes the edge by <1%/yr — the reversion is **not** a continuous-series roll artifact. Direction-agnostic corr: {'COMM_ZS_d.csv': 0.006, 'COMM_ZM_d.csv': -0.143, 'COMM_ZL_d.csv': 0.008} (clean). **Kill criterion: mean net < 2× round-trip cost and Stage-1 gross failure.** Classification per 9.3: structural (b) — the edge is 6–12× below cost and gross insignificant; no honest salvage exists (re-parameterising cannot close a 10× gap).

Hog-corn — feeder must buy corn and sell hogs on the production calendar

IS (2015-09→2014): mean -14.099%/yr, Sharpe -1.05, win 39.7%, t -3.9 — **fails Stage 1 even gross** (gross OOS Sharpe 0.09–0.41, gross t 0.31–1.39, not significant). Permutation p (S2) = 1.0 (expected: real IS result is negative). Walk-forward (S3): Sharpe -0.88, t -3.37. Roll probe: 146 candidate roll days; excluding them changes the edge by <1%/yr — the reversion is **not** a continuous-series roll artifact. Direction-agnostic corr: {'COMM_HE_d.csv': 0.096, 'COMM_ZC_d.csv': 0.105} (clean). **Kill criterion: mean net < 2× round-trip cost and Stage-1 gross failure.** Classification per 9.3: structural (b) — the edge is 6–12× below cost and gross insignificant; no honest salvage exists (re-parameterising cannot close a 10× gap).

Live/Feeder cattle — feedlot fattening margin: buy feeder, sell fed cattle months later

IS (2015-09→2014): mean -13.139%/yr, Sharpe -2.42, win 20.1%, t -8.89 — **fails Stage 1 even gross** (gross OOS Sharpe 0.09–0.41, gross t 0.31–1.39, not significant). Permutation p (S2) = 1.0 (expected: real IS result is negative). Walk-forward (S3): Sharpe -1.67, t -6.35. Roll probe: 115 candidate roll days; excluding them changes the edge by <1%/yr — the reversion is **not** a continuous-series roll artifact. Direction-agnostic corr: {'COMM_LE_d.csv': 0.03, 'COMM_GF_d.csv': -0.045} (clean). **Kill criterion: mean net < 2× round-trip cost and Stage-1 gross failure.** Classification per 9.3: structural (b) — the edge is 6–12× below cost and gross insignificant; no honest salvage exists (re-parameterising cannot close a 10× gap).

3:2:1 Crack — refiner margin: refinery must run crude and sell products continuously

IS (2015-09→2014): mean -6.725%/yr, Sharpe -0.79, win 33.2%, t -2.97 — **fails Stage 1 even gross** (gross OOS Sharpe 0.09–0.41, gross t 0.31–1.39, not significant). Permutation p (S2) = 1.0 (expected: real IS result is negative). Walk-forward (S3): Sharpe -0.48, t -1.81. Roll probe: 85 candidate roll days; excluding them changes the edge by <1%/yr — the reversion is **not** a continuous-series roll artifact. Direction-agnostic corr: {'COMM_CL_d.csv': 0.312, 'COMM_HO_d.csv': 0.021, 'COMM_RB_d.csv': 0.101} (FLAG crack CL 0.30). **Kill criterion: mean net < 2× round-trip cost and Stage-1 gross failure.** Classification per 9.3: structural (b) — the edge is 6–12× below cost and gross insignificant; no honest salvage exists (re-parameterising cannot close a 10× gap).

Track B — Literature search: zero eligible candidates

Searched: SSRN, general web, CME publications. Queries: hedging pressure / normal backwardation; commodity index (Goldman) roll; first-notice-day/delivery effects; pre-hedging; news under-reaction in commodities. 10 papers extracted per 6.3 — every one either overlaps the graveyard (hedging pressure = COT positioning, already dead), requires per-contract / term-structure / intraday data not on the drive (roll front-running, basis reversal, overreaction), or is documented-decayed (Goldman roll order-flow costs down >80% since 2010, Irwin-Sanders-Yan 2022). **No candidate promoted to testing. This is a legitimate, complete result.**

Track C — Platform scan: zero eligible candidates

myfxbook/verifiedinvesting skipped (black-box by definition). QuantConnect supplemental check surfaced no new visible-logic strategy distinct from already-tested families; prior campaign-30 already executed the QC-derived mechanism ideas (witching, LETF decay, window dressing, Russell reconstitution) — all failed the four-stage framework. TradingView/YouTube: no rule set demonstrable to code precision that wasn't a repackaged graveyard family. **Zero promoted.**

Part 9.2 — Volatility prediction module

Composite HAR-RV forecast of next-day |USDJPY return| (rv1/rv5/rv22 + Bollinger-width percentile + |BTC|), IS fit 2016-21, OOS 2022-26 untouched. **Validates:** quintile calibration monotonic in-sample AND out-of-sample (Q5/Q1 = 1.77 / 1.86), Spearman 0.20 / 0.22, OOS R² 0.016 vs naive rv1 -0.81 and rv22 0.014; permutation p < 0.001 on both rho and quintile-spread (1000 runs). **Does not help the drift:** the drift edge is strongest in the high-vol tercile (+0.160%/event, t 3.77 vs 0.33 mid), so inverse-vol sizing throws away the best trades: Sharpe 1.73 → 1.56, ann +19.1% → +15.9%. Correct use: risk management (caps, stops), never a directional signal; re-test per-strategy rather than assume.

Portfolio-level check (survivors at \$150)

Only one active candidate survives everything: the **USDJPY D1 news drift** (forward ledger live, 4 events; backtest OOS 1,201 events, flat Sharpe 1.73, ann +19% on event-return units). No cross-candidate correlation matrix is computable — no second survivor. Dual Momentum remains formally validated (holdout t 4.51, wf p 0.001) but is **retired by your decision** and excluded. Trade frequency: drift triggers ≈ 22/month (well above the 3/month target; the binding constraint is the \$100 platform minimum order and margin — at 30:1, \$150 supports ≈ \$4,500 notional, i.e. roughly 3–4 concurrent minimum-size positions, so the ~22 monthly triggers must be filtered, not all taken). Risk cap: min(\$5, 8.5% equity) per day shared across concurrent positions.

Final verdict and ranked recommendations

Rank	Candidate	Status	Recommendation / caveat
1	USDJPY news drift (D1, next-day continuation)	Only survivor — forward ledger live (4 events)	Use for 30–60 days, run to its 30–60 event kill gate before committing
2	Volatility forecast (HAR-RV composite)	Validated (p<0.001) but no strategy	Use for risk management and as a candidate filter for future tests
—	All four structural spreads	Dead at retail costs	Mechanism real (cointegration stable OOS) but edge 6–12×
—	Tracks B & C (literature/platforms)	Zero eligible candidates	The literature's best mechanism (index-roll front-running) is

Honest cross-track comparison: of four tracks, only the previously-existing candidate survived; the new tracks produced zero tradeable strategies. This is consistent with the project's base rate (~1,300 strategies tested, one survivor). For next time: the highest-value untested ground is (a) per-contract futures data to unlock the roll/term-structure family, and (b) completing the drift's forward kill gate — not more daily-bar backtests on the current data.

Caveats: financing rates assumed (0.01%/day/leg; T212 publishes no history); broker spread 0.30%/leg estimated from earlier measurements (corn 0.27%, gasoline 0.46%); roll-resistance probe is a heuristic (no per-contract data on the drive); free CFD margin for the \$150 account not programmatically confirmable.